

Article title

SmartSense Monitoring: an open-source wireless PT100 node and gateway for surface-temperature monitoring of industrial equipment

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Abstract

SmartSense Monitoring is an open-source wireless PT100 node and gateway for industrial surface-temperature monitoring. Each battery-powered node combines a XIAO ESP32-C6, a three-wire PT100 probe, a MAX31865 front-end and an Ebyte E220 LoRa radio at 433 MHz. The gateway stores a rolling 30-day history locally and serves a Wi-Fi dashboard without a cloud service. A three-node evaluation includes a 32-day pilot deployment and an extended Node B record, totalling 51,338 samples across the individual recording windows. Raw availability was 76.0%, 32.3% and 75.1%; link availability conditioned on inferred gateway-active periods was 97.8–98.4%. Arrival intervals support an inferred PDR of 95.3–96.6%. Five paired temperature readings provide field agreement after offset adjustment, without establishing absolute accuracy. A five-day partial discharge supports estimated autonomy of 177 days by linear extrapolation and 291–307 days using a LiPo-curve model, not demonstrated service lifetime. A temperature rise was later associated with a bearing anomaly during inspection, supporting condition monitoring without validated diagnosis or prediction. Hardware, firmware, software and analysis artifacts are versioned in a public repository; absent primary CSV files and incomplete component identification limit full replication of the evaluation.

Keywords

Open-source hardware; LoRa; PT100; RTD; MAX31865; ESP32-C6; industrial temperature monitoring; condition monitoring

Specifications table

Hardware name	SmartSense Monitoring
Subject area	• Engineering and material science • Educational tools and open source alternatives to existing infrastructure
Hardware type	• Measuring physical properties and in-lab sensors • Field measurements and sensors • Electrical engineering and computer science
Closest commercial analog	Industrial wireless RTD/condition-monitoring loggers such as the Advantech WISE-2410 and commercial PT100 wireless data loggers.
Open source license	Hardware: CERN-OHL-S v2. Firmware and software: MIT License. Documentation: CC-BY 4.0.
Cost of hardware	Itemized total: 627,000 COP (≈ 157 USD at the project conversion assumption of 4,000 COP/USD). Node-specific parts: 149,000 COP (≈ 37 USD), before shared wiring; see Bill of materials.
Source file repository	SmartSense-Monitoring on GitHub ; source-artifact snapshot identified in Design files. No archived DOI is documented.

1. Hardware in context

Continuous temperature monitoring of motors and other industrial equipment helps identify conditions associated with friction, overload, poor ventilation or mechanical degradation. In a manual inspection, an operator obtains only the value at the instant of measurement; if that value is not systematically recorded, the thermal evolution of the equipment cannot be reconstructed or related to its operating and stop cycles.

Wireless sensor networks distribute measurement points without installing communication cabling at each location [1,2,6]. For low-data-rate applications, LoRa provides a long-range, low-power alternative suited to battery-powered nodes, whose range and scalability have been characterised experimentally [3,7,8]. Relevant precedents include LoRaWAN temperature monitoring with a local-server option [9], an open condition-monitoring sensor with vibration and acoustic measurements [10], and the local SD-based HIGROTERM logger [12]. SmartSense contributes the integration of an external three-wire PT100 and MAX31865, point-to-point radio, battery operation and a gateway-hosted daily history and dashboard. Its contribution is this documented platform and pilot evaluation, rather than invention of its constituent technologies or superiority over all existing systems. It supports condition-based maintenance [4,11] but does not implement or validate a predictive-maintenance algorithm.

A PT100 RTD was selected for its established industrial use, three-wire lead compensation with the MAX31865 and availability in a magnetic surface probe suitable for motor housings. The installed probe is identified only as a magnetic three-wire PT100 in the bill of materials: manufacturer, exact reference, tolerance class and rated assembly range are not documented. The generic RTD conversion range therefore cannot be assigned to the complete installed probe, cable or enclosure. The field comparison below characterises agreement in a narrow temperature interval, not the full measurement range or inter-node repeatability.

Compared with proprietary industrial loggers such as the Advantech WISE-2410, SmartSense Monitoring trades integrated vibration sensing for full openness and local-first operation. Table 1 contrasts both systems by capability rather than price alone, making it suitable for pilot plants and educational settings where flexibility and reproducibility matter more than a turnkey platform.

2. Hardware description

SmartSense Monitoring consists of one or more battery-powered sensor nodes and a gateway/visualizer. Each node acquires the surface temperature of a machine with a magnetic PT100 probe and a MAX31865 RTD front-end, estimates its battery state, and transmits the reading over a 433 MHz LoRa link before entering deep sleep. The gateway receives the frames, identifies each node by its MAC address, applies a configurable offset, stores the samples locally and presents them on an integrated touch display and a Wi-Fi web dashboard.

The documented radio configuration uses 9600 bit/s UART (8N1), 2.4 kbit/s air rate, channel 23 in the stated 433 MHz band and fixed addressing. These settings are available in the project connection guide and firmware; the exact radio variant and final pilot transmit power are not verified. A 120 s nominal cycle provides 720 expected samples per node per day, balancing thermal-history resolution with traffic, storage and battery demand for this pilot.

- It provides automatic periodic surface-temperature logging using commercially available modules and an itemized bill of materials.
- It operates fully offline: the gateway serves its own dashboard over a local Wi-Fi access point, so no cloud account or Internet connection is required.
- It preserves a per-sensor daily history, enabling the study of thermal trends, transients and machine on/off cycles that a spot measurement cannot capture.

Feature	SmartSense	WISE-2410
Open hardware	Project schematics and enclosure files, CERN-OHL-S v2	Commercial design; no open hardware release identified
External PT100	Yes, three-wire surface probe	Integrated temperature sensor; no external PT100 specified
Communication	LoRa 433 MHz (point-to-point)	LoRaWAN
Vibration sensing	No	Yes
Local history	microSD or LittleFS; rolling 30-day store	Depends on gateway/application integration
Cloud dependency	None for local dashboard	LoRaWAN gateway/server required; cloud is not inherently mandatory
IP rating	Not demonstrated	IP66 specified
Autonomy evidence	Estimated from 5-day partial discharge: 177 or 291–307 days; 1000 mAh, 120 s cycle	Up to 2 years specified; hourly updates [5]
Software openness	MIT node/gateway firmware and software	WISE Studio utility; open firmware not identified
Cost basis	149,000 COP per node in listed parts ($\approx$ 37 USD); excludes shared wiring	No comparable dated quotation available

Table 1: Functional comparison with Advantech WISE-2410 [5]. WISE-2410 retains advantages in vibration, IP66 protection and specified lifetime. Autonomy uses different update intervals and evidence types; no controlled superiority claim follows. SmartSense cost uses the project BOM and a 4,000 COP/USD conversion assumption.

- It is a modular basis for further work: additional nodes, alternative RTD probes, or extra variables (e.g. current, vibration) can be integrated by reusing the node design.
- Published project schematics and source code support teaching wireless sensor networks, RTD acquisition and local data handling; the purchased modules retain their manufacturer-specific internals.

Design files

The public source-artifact snapshot is [GitHub commit 5641bcd89d9026c39ccc62ce8c738ba5ee37781b](#), reviewed on 15 September 2026. It includes editable KiCad schematics, node/gateway firmware, software, BOM, enclosure files, analysis scripts, figures and licenses. It does not contain the later local thesis and article corrections described here. No Zenodo deposit or DOI is documented in the reviewed repository. The corrected manuscript branch still requires publication as an identified release.

Primary historical and range-test CSV files referenced by the scripts are not included in the reviewed copy; the final pilot configuration and complete PT100/E220 part identification are also absent. Thus the repository supports inspection and reconstruction of the documented design, but does not yet permit exact replication of every reported experiment. Historical results below are retained from project records and figures, not newly recomputed from primary CSVs.

3. Design files summary

Design filename	File type	Open source license	Location of the file
Node schematic (nodo_sensor)	KiCad schematic + PDF/SVG	CERN-OHL-S v2	hardware/kicad/ in identified snapshot
Gateway schematic (gateway)	KiCad schematic + PDF/SVG	CERN-OHL-S v2	hardware/kicad/ in identified snapshot
smartsense.kicad_sym	KiCad symbol library	CERN-OHL-S v2	hardware/kicad/
Node firmware (Nodo_Lora)	Arduino C++ (.ino)	MIT	Firmware directory specified below
Gateway firmware (maestro_Lora_pantalla)	Arduino C++ (.ino) + web assets	MIT	Firmware directory specified below
Bill of materials (BOM.csv)	CSV spreadsheet	CERN-OHL-S v2	hardware/BOM.csv
Enclosure (node, gateway, lids)	STL + zipped 3MF	CERN-OHL-S v2	hardware/carcasa/

The schematics document SPI, UART, switched sensor power and the battery divider. Both firmware folders are under `sensores espnow lora/version con lora y pagina local/`; they contain acquisition, reception, storage and dashboard logic. The BOM and existing PLA enclosure STL/3MF files are in the paths above. These project files do not replace missing probe specifications or a frozen record of the pilot configuration.

Bill of materials

Costs reproduce the project BOM, reviewed on 15 September 2026, in Colombian pesos (COP). USD equivalents use the project's approximate conversion assumption of 1 USD = 4,000 COP, not a verified dated exchange-rate quotation. Supplier invoices and purchase dates are not included. The table covers three nodes and one gateway; prices are historical project entries rather than current market quotations.

4. Bill of materials summary

ID	Component	Qty.	Unit cost (COP)	Total (COP)	Source	Material
U1	Seeed Studio XIAO ESP32-C6 (node)	3	30,000	90,000	Seeed Studio	Semiconductor
U2	ESP32-C6 Touch LCD 1.47 board (gateway)	1	90,000	90,000	Local distributor	Semiconductor
U3	Ebyte E220-433 LoRa module + antenna	4	30,000	120,000	Ebyte	Semiconductor
RT1	Magnetic three-wire PT100 probe	3	30,000	90,000	Local distributor	Metal
U4	MAX31865 RTD front-end module	3	14,000	42,000	Sigma Electrónica	Semiconductor
BT1	LiPo battery 3.7 V 2000 mAh	3	30,000	90,000	Didácticas Electrónicas	Other
U5	LiPo charge and protection module (TP4056 or equiv.)	3	5,000	15,000	Local distributor	Semiconductor
R1	Resistor 344.8 k Ω (battery divider)	3	–	–	Local distributor	Other
R2	Resistor 994.3 k Ω (battery divider)	3	–	–	Local distributor	Other
MEC1	3D-printed PLA enclosure	4	10,000	40,000	Self-printed	Polymer
W1	Wires, connectors, headers, hardware, adhesives	1	50,000	50,000	Local distributor	Other

Itemized total: 627,000 COP ($\approx$ 157 USD at the stated conversion). Node-specific parts sum to 149,000 COP per node, including the 2000 mAh cell and enclosure; gateway board, radio and enclosure sum to 130,000 COP. Shared wiring/hardware contributes 50,000 COP, including battery-divider resistors. These entries do not establish a labor, shipping or certification cost. The battery-characterization test instead used a 1000 mAh cell.

5. Build instructions

The following steps describe the construction of one sensor node and the gateway. Refer to the node and gateway schematics (Design files) and to the Bill of materials for component designers.

Sensor node

1. Use a three-wire magnetic PT100 probe and document its manufacturer, reference, tolerance and rated assembly temperature before replication; these fields are missing for the pilot probe. Configure the MAX31865 module in three-wire mode (bridge the 2/3-wire jumper), and connect the PT100 leads: the two leads from the same probe end to F+ and RTD+, and the remaining lead to RTD-, bridged to F-.
2. Wire the MAX31865 SPI bus to the XIAO ESP32-C6: CLK $\rightarrow$ D6 (GPIO16), SDO $\rightarrow$ D5 (GPIO23), SDI $\rightarrow$ D4 (GPIO22), CS $\rightarrow$ D3 (GPIO21). Power the module VIN from D2 (GPIO2) so the firmware can switch it off during deep sleep.

3. Connect the E220 LoRa module by UART: E220 TXD $\rightarrow$ D7 (GPIO17), RXD $\rightarrow$ D8 (GPIO19), M0 $\rightarrow$ D10 (GPIO18), M1 $\rightarrow$ D9 (GPIO20); power VCC/GND from 3V3/GND.
4. Build the battery divider: VBAT – R1 (344.8 k Ω) – A0 (GPIO0) – R2 (994.3 k Ω) – GND. Connect the LiPo cell through the TP4056 charge/protection module to the XIAO battery pads.
5. Flash the node firmware (`Nodo_Lora.ino`) with the Arduino IDE and the ESP32 board package. Install the Adafruit_MAX31865 library. Set RREF to the measured reference resistance of your MAX31865 module (421.1 Ω in this build).
6. Fit the components in the 3D-printed PLA enclosure and attach the PT100 magnetically to the target surface.

Gateway

1. Connect the E220 module to the ESP32-C6 Touch LCD board: E220 TXD $\rightarrow$ GPIO8 (gateway RX), RXD $\rightarrow$ GPIO7 (gateway TX), and both M0 and M1 to GPIO9; power VCC/GND from 3V3/GND.
2. Optionally wire a critical-alarm output (LED, buzzer or relay driver) to GPIO16.
3. Flash the gateway firmware (`maestro_Lora_pantalla.ino`) and upload the web dashboard assets to the on-board storage (LittleFS). Insert a microSD card for the history.

Safety. Use a fused LiPo cell with the TP4056 protection module and observe LiPo handling precautions. Ensure the PT100 and enclosure are rated for the surface temperature of the monitored equipment and mounted so they do not interfere with moving parts.



Figure 1: Assembled sensor node mounted on a motor housing: the PT100 probe with the MAX31865 module (left) and the node enclosure with the LoRa antenna (right).

6. Operation instructions

1. Power the gateway. It creates a Wi-Fi access point (`Sensores_Temp`) and serves the dashboard at <http://192.168.4.1>.
2. Power each node. On first boot it transmits periodically (default configurable between 5 and 7200 s; 120 s in the pilot). The gateway registers each node by its MAC address.
3. In the dashboard, assign names to sensors, group them by machine, set the sampling interval, apply a per-node temperature offset, and configure warning and critical thresholds.
4. Monitor the live values (temperature, battery, connectivity) and consult the per-day 24-hour history, including minimum, maximum and average, and compare sensors of the same machine.
5. Preserve copies of the per-sensor stored history before the 30-day rotation removes older dates. The project analysed exported CSVs and the software replica can import them; direct web CSV download is not verified in the reviewed gateway firmware and is not claimed as a demonstrated interface function.

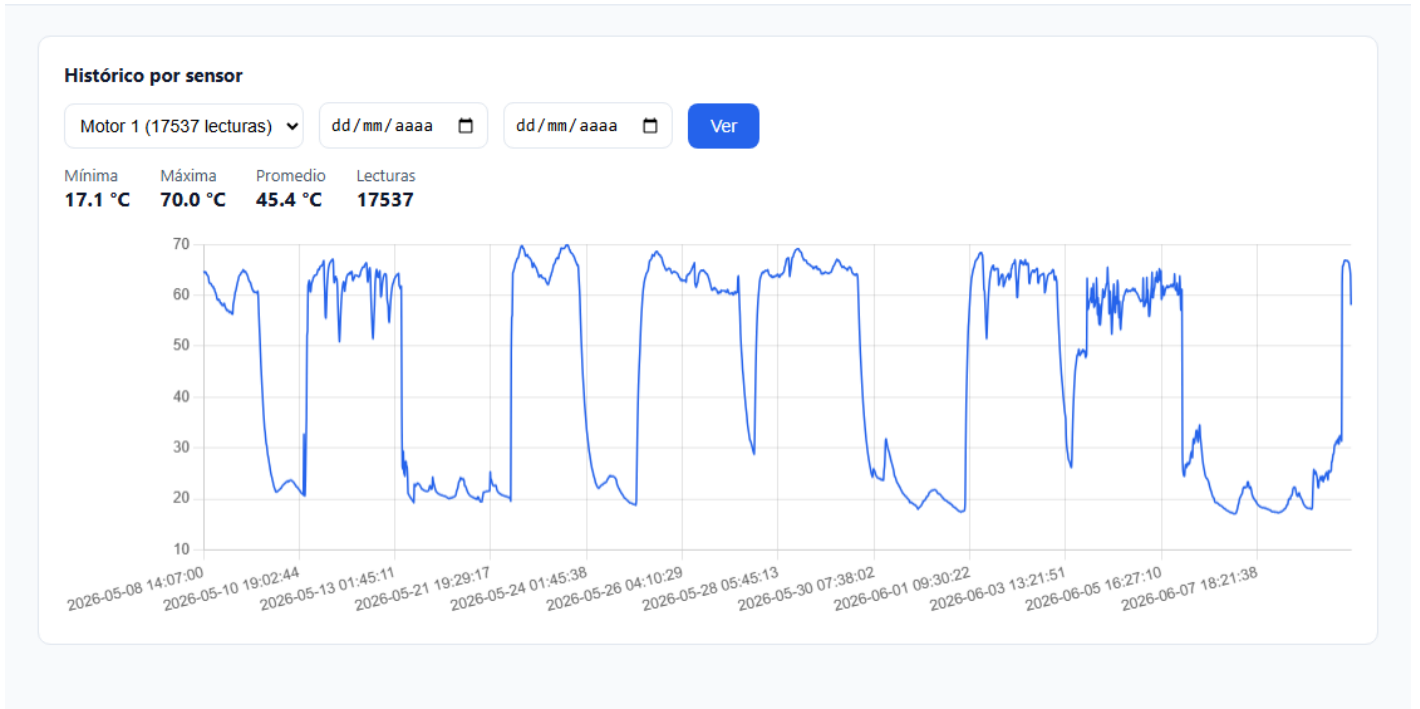


Figure 2: Local web dashboard: per-sensor temperature history over the deployment, with minimum, maximum and mean statistics. Shown here running on the open-source software replica with generic sensor labels.

Safety. The critical-alarm output is a supervisory aid, not a safety interlock; do not use it as the sole protection for equipment or personnel.

7. Validation and characterization

The three-node evaluation includes a 32-day pilot deployment (8 May–9 June 2026 for Nodes A and C), with Node B recorded from 10 May to 20 July 2026. The 51,338 samples are the sum of these individual files, not a count over 32 complete days common to all three nodes. Figures comparing nodes use their overlapping interval. The nominal cycle was 120 s and the documented median inter-arrival interval was 121 s. Separate partial-discharge and radio-range tests were performed in August.

Availability. Table 2 separates raw system-level data availability from link availability conditioned on inferred gateway-active periods. The generator `analisis/figuras_validacion.py` uses 121 s as the expected cadence. Raw availability divides received samples by the expected count across each node’s first-to-last-record span. Conditioned availability removes the entire duration of each inter-arrival gap exceeding 363 s before computing the expected count. Simultaneous gaps in Fig. 8 support a common-cause interpretation, including gateway power interruptions during plant stoppages; they do not independently prove the power state or distinguish reception, clock and storage failures. Thus 97.8–98.4% describes classified active periods, not overall system availability, and the excluded hours cannot all be asserted to be confirmed power-off hours.

Reception cadence and inferred packet delivery. Retained short intervals cluster near a 121 s median (Fig. 9). The separate generator `analisis/perdida_paquetes.py` uses a 120 s nominal cycle and excludes intervals longer than 1800 s. For retained gaps it rounds the interval divided by 120 s to k , infers $k - 1$ missing frames when $k \geq 2$, and computes received/(received + inferred missing). The documented inferred PDR is 96.55%, 96.07% and 95.32% for A, B and C, respectively: 95.3–96.6% rounded, with inferred loss of 3.5–4.7%.

Node	Samples	Raw avail.	Conditioned link avail.	Gaps	Hours excluded
Node A	17,537	76.0%	98.4%	66	176
Node B	16,487	32.3%	98.3%	73	1154
Node C	17,314	75.1%	97.8%	81	180

Table 2: Documented availability in the individual recording windows. Node B’s longer file includes a gap of approximately 33 days; its raw availability is not a comparison over the same calendar span as A and C. Gap counts and excluded hours are temporal classifications, not a direct power log.

Each estimate uses its individual node file, including Node B’s extension. These filters and denominators differ from availability; neither metric is a direct sent-versus-received count. Without a historical sequence counter and send log, cadence assumptions cannot distinguish radio loss from absent acquisition or persistence.

Representative 60-minute window. The historical window of 10 May 2026, 00:00–01:00 (Fig. 4), contains 29, 30 and 30 records for A, B and C against 30 nominally expected at a 120 s cycle: received/expected ratios of 96.7%, 100% and 100%. The denominator is a schedule, not an observed transmission count. The script selects a window with at least 25 records per node, so this is a coverage-selected illustration, not an independent test or an unbiased estimate for all hours. RSSI was enabled later for the range test; the interface used did not expose SNR. A sequence counter added after the deployment supports future direct PDR testing but cannot recover historical sends. End-to-end latency and service recovery require synchronized node/gateway measurements.

Range test. On 20 August 2026 a node moved away from the gateway and back in an urban non-line-of-sight path with buildings in between (Fig. 3). Valid packets were received at the farthest observed distance, approximately 688 m, with RSSI near -98 dBm at 20:44–20:54, compared with approximately -15 dBm nearby. The test ended voluntarily with the link still operational: at least 688 m is a demonstrated lower bound in this geometry. No numerical sensitivity margin is claimed because the exact E220 variant and final physical configuration are incompletely documented. The recorded battery indicator stayed near 94% during 2.6 h; this voltage-derived indication does not measure charge consumption. Motion and changing obstacles prevent a controlled RSSI–distance characterization; fixed positions and repeated trials remain future work.

Temperature field comparison. The purpose was to assess consistency of the installed node with a field reference on the suction-fan motor. Five simultaneous paired readings were recorded near the same surface point in the 65–71 °C interval against a Fluke 62 MAX infrared thermometer (Table 3). Their mean absolute difference is 0.06 °C (approximately 0.1 °C rounded) and their maximum absolute difference is 0.1 °C; the signed mean is 0.02 °C. The project summary separately reports a maximum of 0.4 °C. That reported value is retained as a provenance limitation: it cannot be reconstructed from the five transcribed pairs, and resolving it requires the original measurement record.

Time	SmartSense (°C)	Fluke (°C)	$ \Delta T $ (°C)
06:40	71.0	71.0	0.0
07:06	70.6	70.6	0.0
07:58	69.7	69.6	0.1
08:09	69.5	69.4	0.1
09:49	65.1	65.2	0.1

Table 3: Five transcribed field-comparison pairs for one node. The maximum of these rows differs from the 0.4 °C maximum communicated in the project summary.

The FLIR image (Fig. 5), attributed in the field record to a C8 camera, corroborates the location and approximate 70–71 °C magnitude of the hotspot; it is not documented as synchronized to every paired reading. The result is field consistency, not traceable calibration. The Fluke was also used to adjust the offset, so the compar-

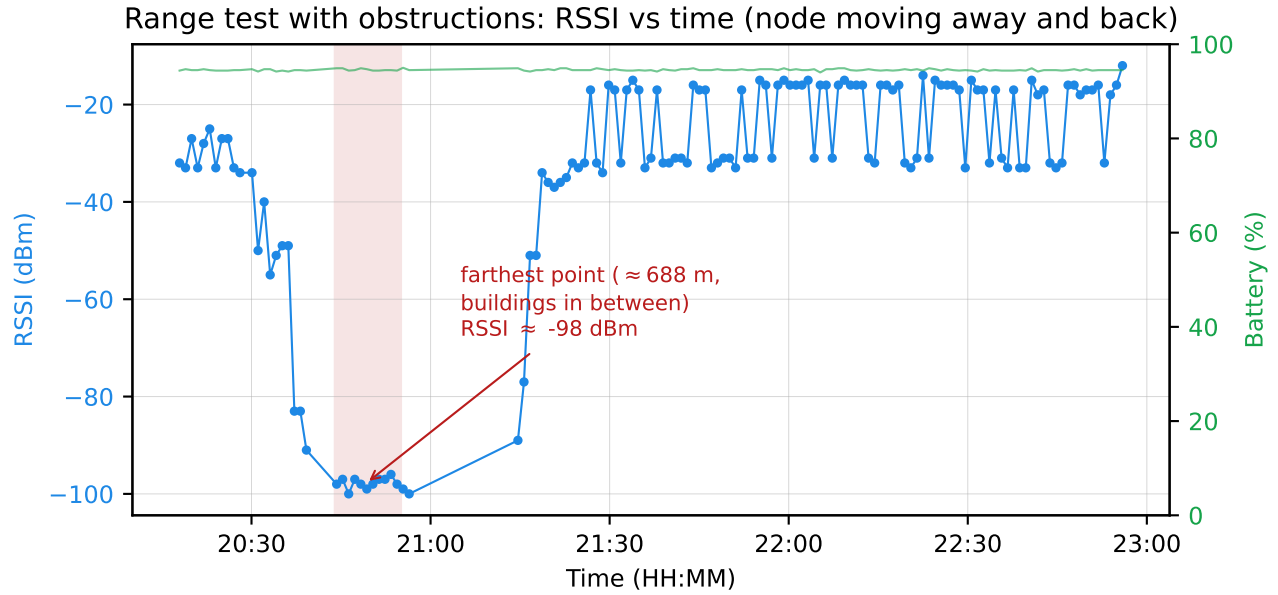


Figure 3: Urban obstructed-path test of 20 August 2026: per-packet RSSI (blue) and voltage-derived battery indication (green) against time during a mobile outward and return route. The link remained operational near 688 m at approximately -98 dBm; distance was not a controlled fixed-position sweep.

ison is not independent. The project cites instrument specifications of ± 1.5 °C for the Fluke and ± 2 °C for the FLIR, with no current Fluke calibration certificate. These limits exceed the order of the 1 °C target and are not a combined uncertainty budget for the installed system. Emissivity and viewed area affect infrared readings; contact and mounting affect the PT100. Complementary characterization must use a calibrated contact reference, separate adjustment from verification, compare the three nodes at multiple temperatures and account for reference, conversion, lead, mounting and repeatability contributions. Absolute accuracy and inter-node repeatability remain unestablished.

Field observation (case study). The suction-fan motor history includes rises and falls and a recorded maximum of 70.0 °C (Fig. 7). Plant observations associated rising segments with periods when the machine stopped while the fan continued spinning unloaded; no synchronized machine-state log establishes the timing or lag of each transition. The thermal image then provides spatial context for the hotspot (Fig. 5). Subsequent inspection associated the case with a bearing anomaly. This sequence—thermal history, operational context, thermal image and later inspection association—illustrates the use of historical data in condition monitoring. It does not establish diagnosis, prediction, lead time, causality, sensitivity or false-alarm performance; load, ventilation, mounting and other factors were not controlled.

Instrumented activity and temperature range. A firmware-instrumented interval was 2.279 s in eight of ten documented starts, with small variations reported for the remaining two. Instrumentation begins after initial serial setup and delay, so this is not total physical power-on-to-service time or synchronized network latency. Dividing by the 120 s cycle gives approximately 1.9% instrumented duty cycle; complete reset recovery remains to be measured through reception and persistence at the gateway. Across the individual historical files, surface temperature spanned 15.3–70.0 °C, with per-node means of 39.5–45.4 °C. These statistics describe recorded thermal behaviour, not calibrated range or measurement uncertainty.

Estimated battery autonomy. A controlled five-day partial discharge under the nominal 120 s operating cycle used a 1000 mAh cell from 13 to 18 August 2026. Two voltage readings, 4.112 V and 4.089 V over 120 h, give a slope of approximately 0.19 mV/h. The generator `analysis/autonomia_bateria.py` extrapolates that slope to 3.30 V for a conservative linear estimate of 177 days. Its assumed typical LiPo open-circuit-voltage/state-of-charge curve, fitted to the same two points, gives 291–307 days at cutoffs of 3.30 and 3.00 V, respectively

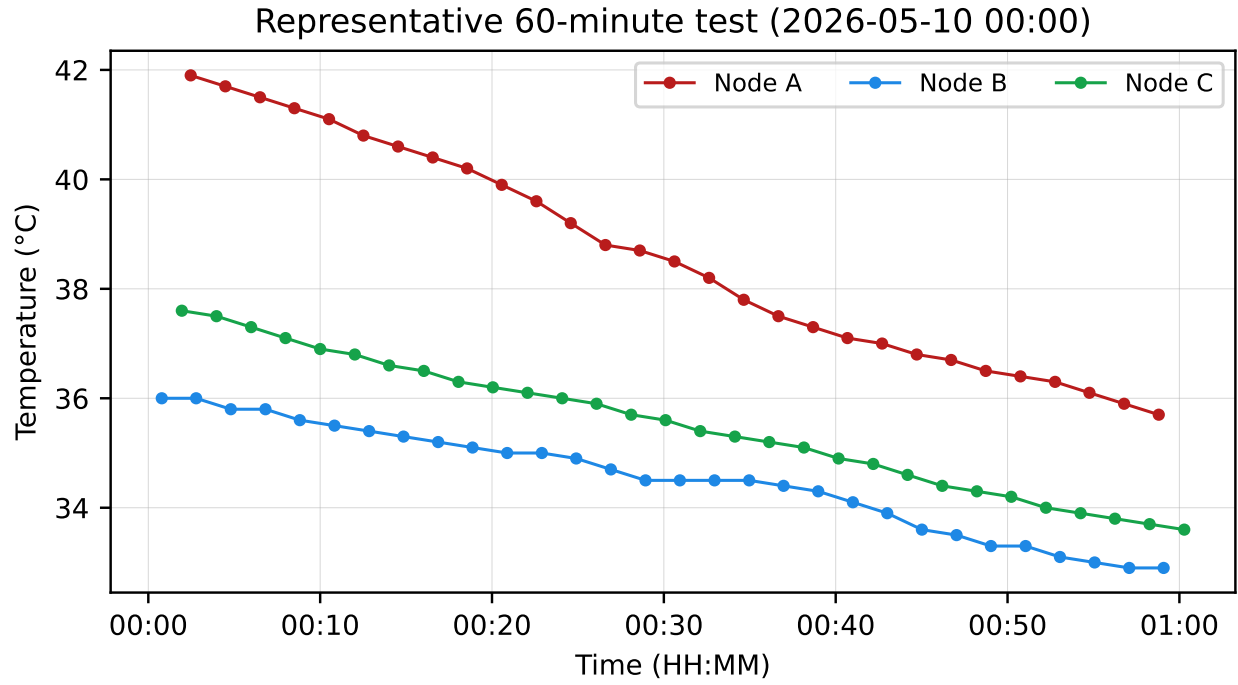


Figure 4: Temperature during a coverage-selected historical hour with three nodes. The original plot label “test” identifies this illustrative window, not a separate trial with a transmission counter.

(Fig. 6). This range expresses two cutoffs under one model, not a statistical confidence interval. The curve is a project modeling assumption without a documented primary cell characterization.

Both values are estimated autonomy, not experimentally demonstrated lifetime. The linear result is conservative relative to the LiPo model but is not a guaranteed physical lower bound. Temperature, load, relaxation, ADC resolution, aging and self-discharge can alter the projection from this small voltage drop. Model-implied average current is approximately 0.13–0.24 mA; neither it nor the assumed microcontroller sleep current is an integrated measurement of the complete node. The 1.9% instrumented duty cycle does not establish energy consumption. A full discharge to a specified cutoff and current logging by operating state are needed; no automatic doubling is claimed for the 2000 mAh BOM cell.

Capabilities and limitations.

- Acquires periodic surface-temperature readings with a three-wire PT100 and configurable per-node offset; probe class, rated assembly range and full part identification remain undocumented.
- Raw availability is 76.0%, 32.3% and 75.1% in the individual windows; conditioned link availability is 97.8–98.4% and PDR is inferred at 95.3–96.6%, using different gap filters.
- Operates locally with a rolling 30-day per-sensor store; older dates are removed after rotation. Pre-rotation exports preserved the records analysed beyond 30 days, including Node B’s longer history. LittleFS is an alternative storage medium, not a mirrored backup. Direct web CSV export is unverified and primary CSVs are absent from the reviewed repository.
- Five pairs support field agreement after adjustment to the same Fluke reference. The project-reported 0.4 °C maximum is not reconstructible from those pairs. Absolute accuracy requires an independent calibrated contact reference and an uncertainty budget.
- A five-day partial discharge supports estimated autonomy of 177 days or 291–307 days under the stated model and cutoff assumptions; extended lifetime and complete-node current consumption were not experimentally demonstrated.
- LoRa range of at least ≈ 688 m in a non-line-of-sight urban path, with the link operating down to $\approx$



Figure 5: Thermal image of the suction-fan motor, attributed to a FLIR C8 in the field record, locating a hotspot near 70–71 °C. It supplies spatial context without proving a bearing cause or the PT100’s absolute accuracy.

- 98 dBm RSSI (test stopped voluntarily, not by link failure).
- The exact E220 variant and final pilot transmit power are not verified. The interface used did not expose SNR, and synchronized end-to-end latency and complete service recovery were not measured. Three deployed nodes do not constitute controlled inter-node repeatability or a network-capacity test.

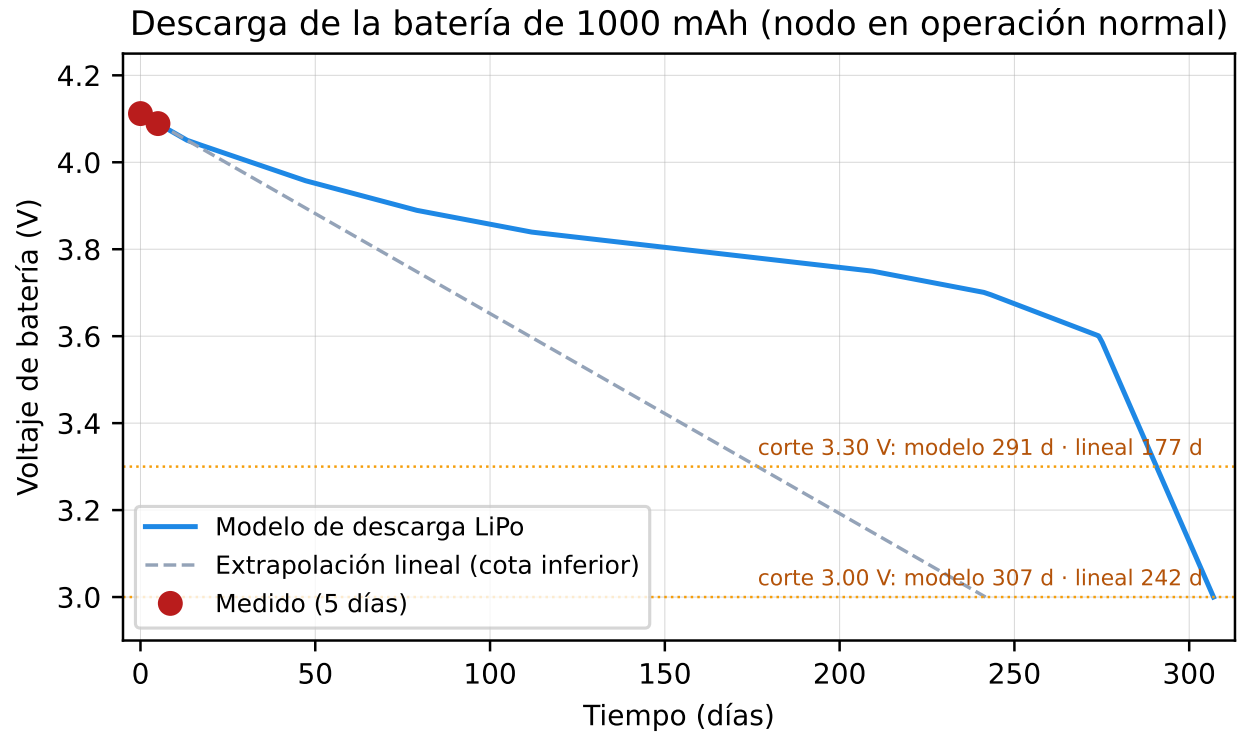


Figure 6: Two measured voltage points over five days with a 1000 mAh cell and extrapolated curves. Estimated autonomy is 177 days (linear, 3.30 V cutoff) or 291–307 days (assumed LiPo curve, 3.30/3.00 V cutoffs). The original plot’s lower-bound label compares models; it does not guarantee service lifetime.

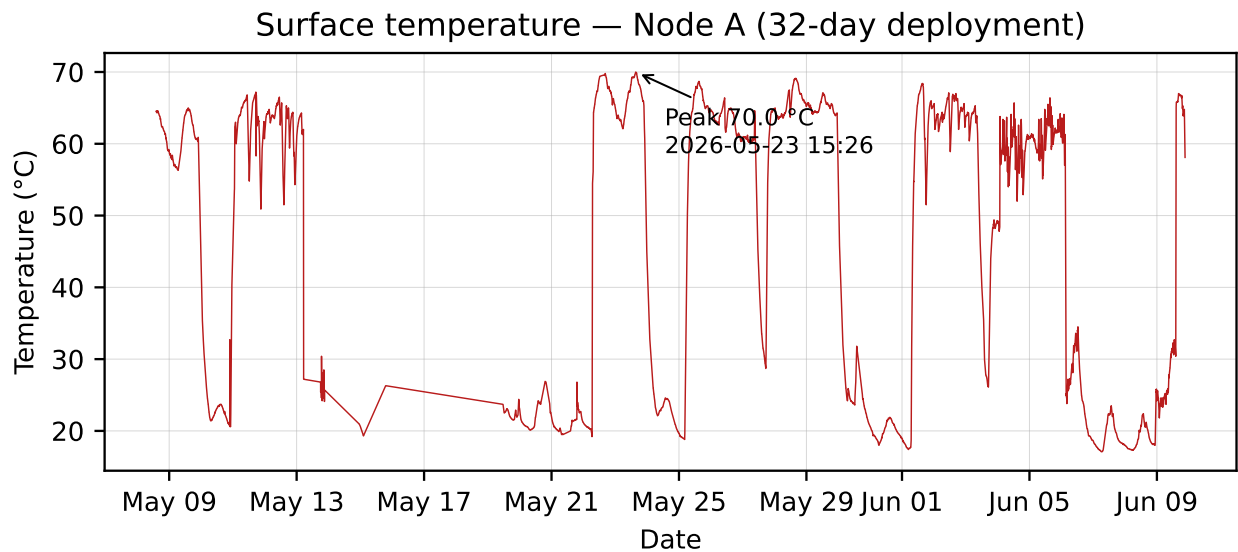


Figure 7: Node A surface-temperature history during the 32-day pilot deployment. The 70.0 °C peak is a historical observation; its annotation does not identify a mechanical cause.

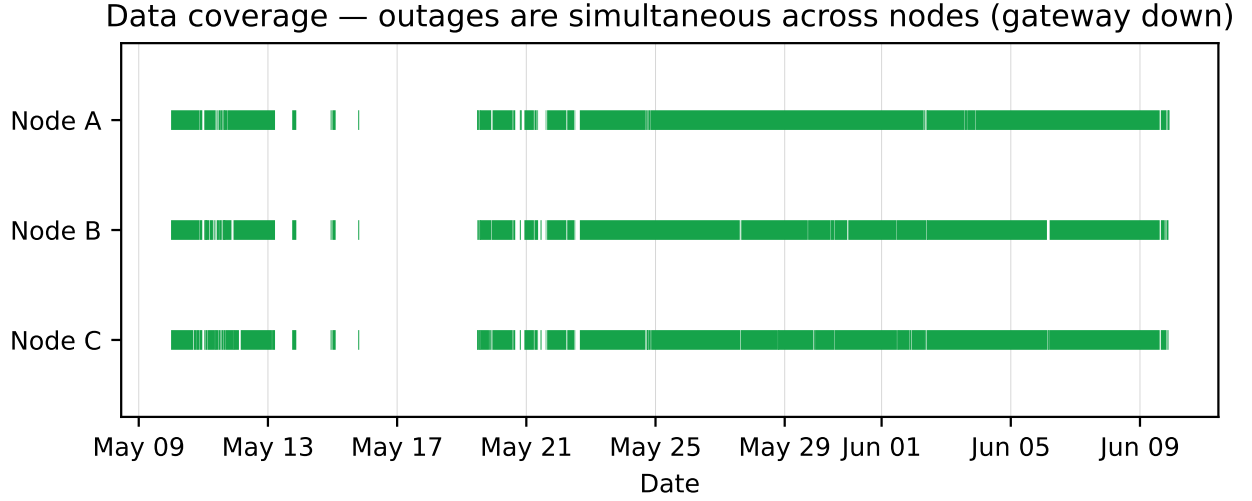


Figure 8: Data coverage in the three-node overlapping interval. White gaps retain missing data. The original “gateway down” plot label is an interpretation compatible with simultaneous interruptions, not an independently verified power-state record.

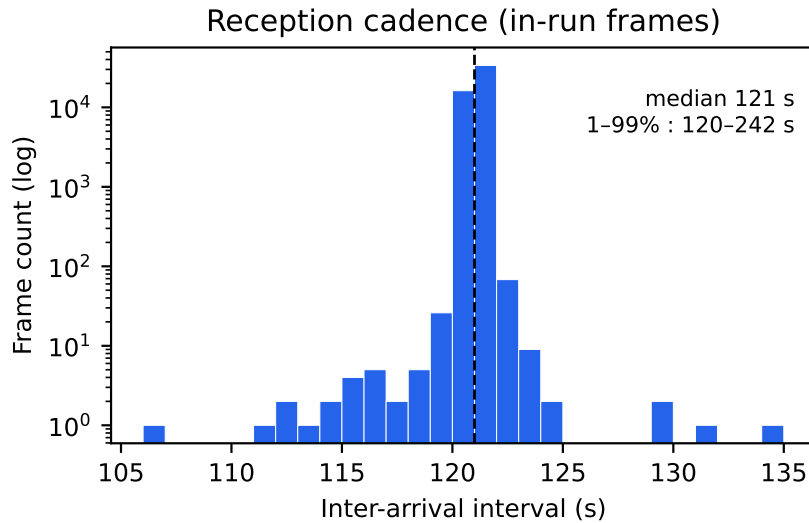


Figure 9: Short inter-arrival intervals retained after excluding gaps longer than 363 s, with logarithmic frame-count axis. The plotted range concentrates around the 121 s median and does not represent the full outage distribution.

Ethics statements

This work did not involve human subjects or animal experiments.

CRediT author statement

Jesus Andre Bojaca Lopez: Conceptualization, Methodology, Software, Investigation, Writing – Original draft. **David Santiago Puentes Cardenas:** Conceptualization, Hardware, Software, Validation, Data curation, Writing – Review & Editing. **Jhon Edwin Vera:** Supervision. **Sergio Francisco Mora Martinez:** Supervision.

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